

Reproduction of yandex-research/tabm: TabM on California housing, 15 seeds

K-Veritas Team

Submitted by K-Veritas Team

kv:2610.00021v1 · Published 2026-10-08

Independent reproduction · Artifact evaluation · Machine learning

Abstract

Re-ran TabM on the California housing dataset at the pinned commit with the paper's 15 tuned evaluation configs (paper/exp/tabm/california/0-evaluation, seeds 0 to 14), and compared each test RMSE with the report stored in the repository for the same seed. Environment: PyTorch 2.11 with CUDA 12.8 in place of the pinned 2.0.1 (the GPU needs it), other packages at the versions in pixi.toml, TORCH_FORCE_NO_WEIGHTS_ONLY_LOAD=1 for the checkpoint format. Datasets and stored results for other experiments were moved out of the working copy; Python's hostname lookup returns localhost so it does not appear in log file names. One NVIDIA RTX 5060 Ti.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

This cover page is not signed. Check it against the SHA-256 listed at <https://kveritas.org/records/2610.00021v1>, and verify each report with its code bundle: `kveritas verify <report.pdf> --bundle <bundle.zip>`.

1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	TabM California, 15 seeds	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: TabM California, 15 seeds

Run	Metric	Final value	Step
1	test_rmse	0.443899	14
1	paper_test_rmse	0.44022	14
1	test_rmse_std	0.00136971	

Run	Inline claim	Value	Phase
1	test_rmse_mean	0.441733	
1	paper_test_rmse_mean	0.441392	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	1635.8 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	ba56df73ac77d5ddcc0f2725815cccb180f65040647023f1ab208366e98f30f9	6.9 MB
r1.kvbundle.zip	84a715982fed380fe7bb24beef460e16d65c94a270263a1019a3c8deba654621	192 KB

5 Links and licence

Paper: <https://arxiv.org/abs/2410.24210>

Reproduction of: <https://github.com/yandex-research/tabm/tree/28e47ae301c92ec37787dde1ce923a0793f405b4>

Record: CC-BY 4.0. Code: Apache-2.0, as declared by the authors. Tags: tabular, ensembles, tabm, california-housing.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (TabM California, 15 seeds), kv:2610.00021v1/r1. Data hash d1c525f2d120c4578b84e3f423dcc756e3d53f5a359fbecf3f573cb066a7edf.
<https://kveritas.org/records/2610.00021v1/r1.pdf>